IES FRANCISCO DE GOYA

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Inglés Técnico:
GPU Comparison: NVIDIA GeForce RTX 5090 vs AMD Radeon RX 9900 XTX



1. Overview and Technical Specifications

Feature	NVIDIA GeForce RTX 5090	AMD Radeon RX 9900 XTX
VRAM Memory	32 GB GDDR7	24 GB GDDR6X
Power Consumption	600 W	450 W
Target Resolution	4K / 8K Gaming	4K Gaming
Ray Tracing Performance	Extremely High	High
Price (MSRP)	$1,999	$1,199

Choosing a high-end graphics card is always a difficult choice because both brands offer strong hardware. In this comparison, we will analyze five key features: design, technical specifications, performance, power consumption/durability, and price to see which card is the best option for users today.

2. Feature Comparison

1. Design and Size

The RTX 5090 is noticeably bigger and heavier than the RX 9900 XTX. NVIDIA's card takes up almost four slots in a PC case, so you need a very large tower. On the other hand, AMD’s design is more compact and easier to install in standard computer cases.

2. Technical Specifications

When looking at the numbers, NVIDIA offers more VRAM with 32 GB of modern GDDR7 memory, whereas AMD comes with 24 GB of GDDR6X. This means the RTX 5090 has a wider memory bandwidth, making it better prepared for future heavy workloads.

3. Speed and Performance

In pure power and frame rates, the RTX 5090 is clearly faster than the RX 9900 XTX, especially in ray-tracing technology and AI rendering (DLSS). However, AMD's card is still extremely powerful for traditional 4K gaming and runs fast enough for almost any modern video game.

4. Power Consumption and Durability

The RX 9900 XTX is much more efficient than its competitor. The RTX 5090 consumes up to 600W and gets very hot, which means you need a stronger power supply and a top-tier cooling system to keep it durable over time. The AMD model is less demanding and easier to keep cool.

5. Price

Price is the biggest difference here. The RTX 5090 costs around $1,999, making it much more expensive than the RX 9900 XTX, which costs around $1,199. NVIDIA delivers better performance, but AMD offers a far better balance between price and quality.

3. Final Choice and Justification

After comparing both graphics cards, I would choose the **AMD Radeon RX 9900 XTX**. Although NVIDIA's RTX 5090 is undeniably the most powerful card on the market, it is far too expensive for most users and consumes a huge amount of electricity. The RX 9900 XTX offers fantastic 4K gaming performance, uses less power, and costs almost $800 less, making it a much smarter investment overall.

References

- TechPowerUp GPU Database: https://www.techpowerup.com
- Tom's Hardware GPU Benchmarks & Hierarchy: https://www.tomshardware.com